

Application Specification: Post-Operative Investigation Follow-Up Platform

1. Document Overview

Field	Value
Document title	Post-Operative Investigation Flow Chart — Digital Follow-Up Platform
Version	1.6 (Draft)
Date	August 23, 2026
Source inputs	Google Doc ("Automate follow up"), sample flowchart (pediatric post-op liver patient), Requirement Update: Conversational Chat & Multi-Doctor Collaboration/Case Transfer

2. Problem Statement

Remote patients recovering from major surgeries (specifically **Post-Operative Liver Transplant** and chronic liver disease face significant challenges tracking recurring lab reports, liver enzyme levels, immunosuppressant drug trough levels (Tacrolimus, Everolimus), vital signs, and complex medication regimens in paper charts or spreadsheets, sharing updates via unorganized email or WhatsApp.

Surgeons and hepatologists need to review longitudinal lab trends, adjust medication dosages (often annotating changes in red ink), request targeted follow-up tests, and collaborate with co-managing doctors and specialists. Additionally, patient care extends over years or decades, requiring seamless handoffs between primary transplant surgeons and consulting hepatologists or secondary specialists.

Remote patients currently track post-operative lab results and immunosuppressant medication doses in a **paper flowchart**, then share updates with their doctor via **email or WhatsApp**. The doctor reviews the chart, adjusts medication doses (often annotating changes in a different color), and replies with instructions. The patient manually updates the chart and repeats the cycle at the next follow-up.

Furthermore, between periodic lab updates, patients lack a secure, audited channel to ask quick questions or report emerging symptoms, forcing reliance on unorganized WhatsApp chats. Additionally, post-operative transplant care frequently involves multiple clinicians (surgeons, hepatologists, pediatricians). Currently, there is no structured way for a primary doctor to grant secondary doctors access to a patient's chart or formally transfer primary case ownership when duty shifts or clinical handoffs occur.

This workflow is error-prone, hard to audit, fragmented across un-encrypted channels, and inefficient for both patients and care teams. The goal is to **digitize the flowchart, automate the follow-up loop, provide integrated clinical chat, and enable multi-doctor collaboration and case transfer** while preserving the familiar tabular format and clinical semantics.

While **Release 1 is dedicated to Post-Operative Liver Transplant care**, the underlying platform is built on an **extensible, protocol-driven architecture** that allows seamless scaling to other long-term chronic conditions (such as Renal Transplant, Chronic Kidney Disease, Heart Failure, and Diabetes) without core codebase refactoring.

3. Goals & Success Criteria

Primary goals

1. Replace the paper flowchart with a structured digital record.
2. Let patients enter lab values and current medication doses remotely.
3. Let doctors review entries, prescribe/adjust doses, and respond in one place.
4. Deliver doctor responses back to the patient automatically (replacing manual email/WhatsApp for the core loop).
5. **Provide secure, integrated conversational chat** between the patient/caregiver and authorized care team doctors, complete with media attachments, voice notes, and clinical event linking.
6. **Support multi-doctor participation**, enabling the primary doctor (or patient) to authorize secondary/consulting doctors with granular permissions (co-managing vs. view-only consult).
7. **Enable formal patient case transfer** allowing the primary doctor to transfer primary case ownership to another clinician with a full handoff audit trail.
8. Maintain a longitudinal history of labs, drug levels, doses, dose changes, chat logs, and doctor access grants.
9. Send automated **SMS and WhatsApp reminders** for upcoming follow-up milestones so patients do not miss lab submissions.

Success criteria

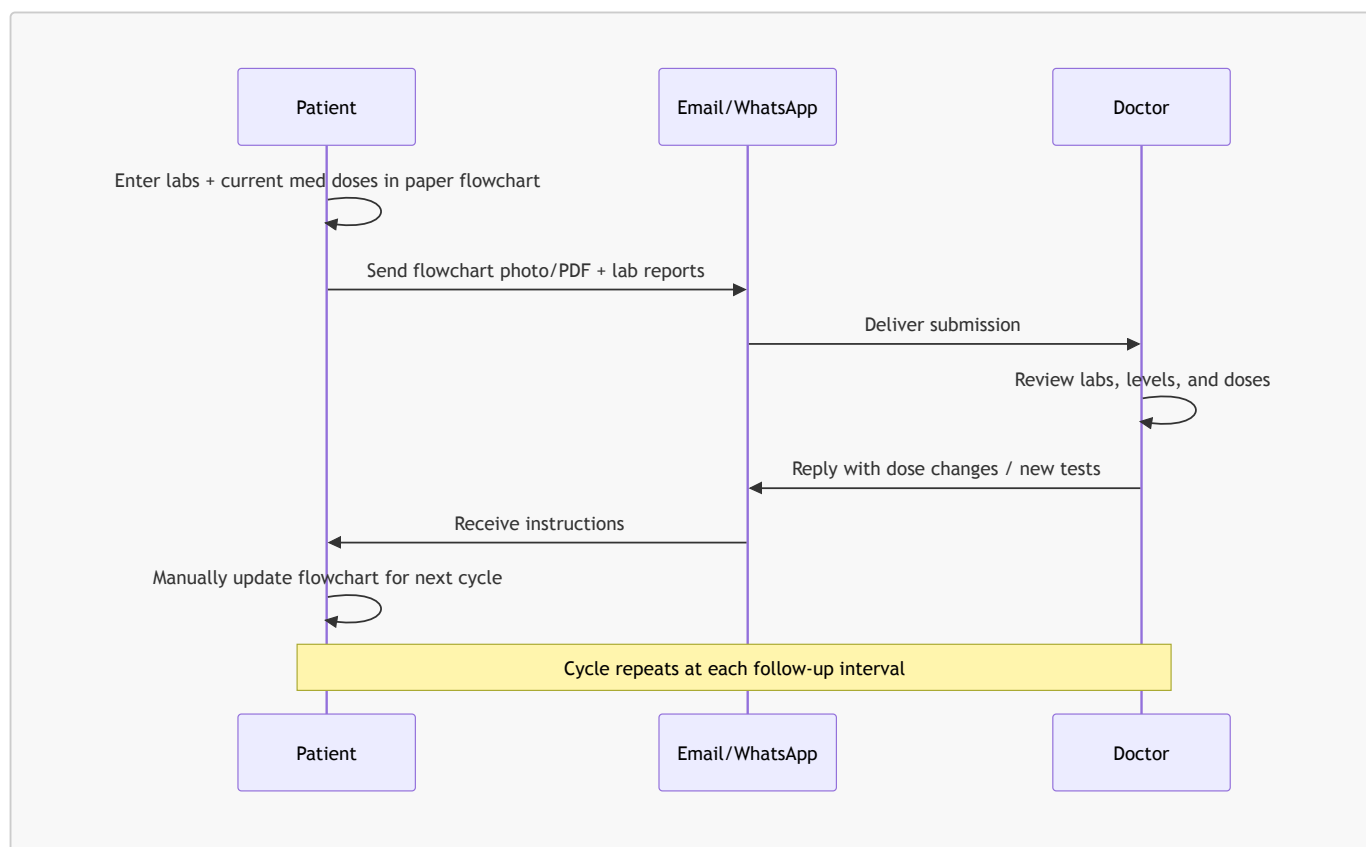
- A patient can submit a new follow-up row in under 5 minutes on mobile.
- A doctor can review a submission and send a dose adjustment in under 3 minutes.
- Every dose change, chat message, authorization grant, and case transfer is fully traceable (who, when, prior vs new state).
- Primary doctors can grant consulting access or transfer a patient case to another doctor in under 1 minute with instant permission updates.
- Patients and doctors can send and view contextual chat messages directly tied to flowchart submissions and lab trends.
- The digital chart mirrors the paper layout closely enough that existing users can adopt it without retraining.

4. Users & Personas

Role	Description	Primary needs
Patient / Caregiver	Remote post-op patient (or parent/guardian for pediatric cases)	Simple data entry, view doctor instructions, upload lab reports, message care team, view active care team doctors
Primary Doctor	Lead transplant/hepatology surgeon or consultant owning the case	Review trends, enter/adjust doses, request tests, reply to patient, chat with patient, authorize co-managing/consulting doctors, transfer case ownership
Co-Managing Doctor	Authorized care team doctor (e.g. Dr. Rajesh Dey, Dr. Tejai B, Dr. Barun) with edit/prescribe access	Review patient chart, co-prescribe doses, participate in patient chat, enter clinical notes
Consulting Doctor	Specialist called in for specific advice (e.g. Nephrologist, Infectious	View-only chart access, review labs, add clinical recommendations/notes in consult thread

Role	Description	Primary needs
	Disease specialist)	
Transferred (Former) Doctor	Doctor who previously owned the patient case	Read-only audit access to historical entries recorded during their tenure as primary doctor
Admin	Clinic/hospital staff	Onboard patients, manage doctor accounts, execute administrative case transfers when required

5. Current Workflow (As-Is)



Observed from sample flowchart:

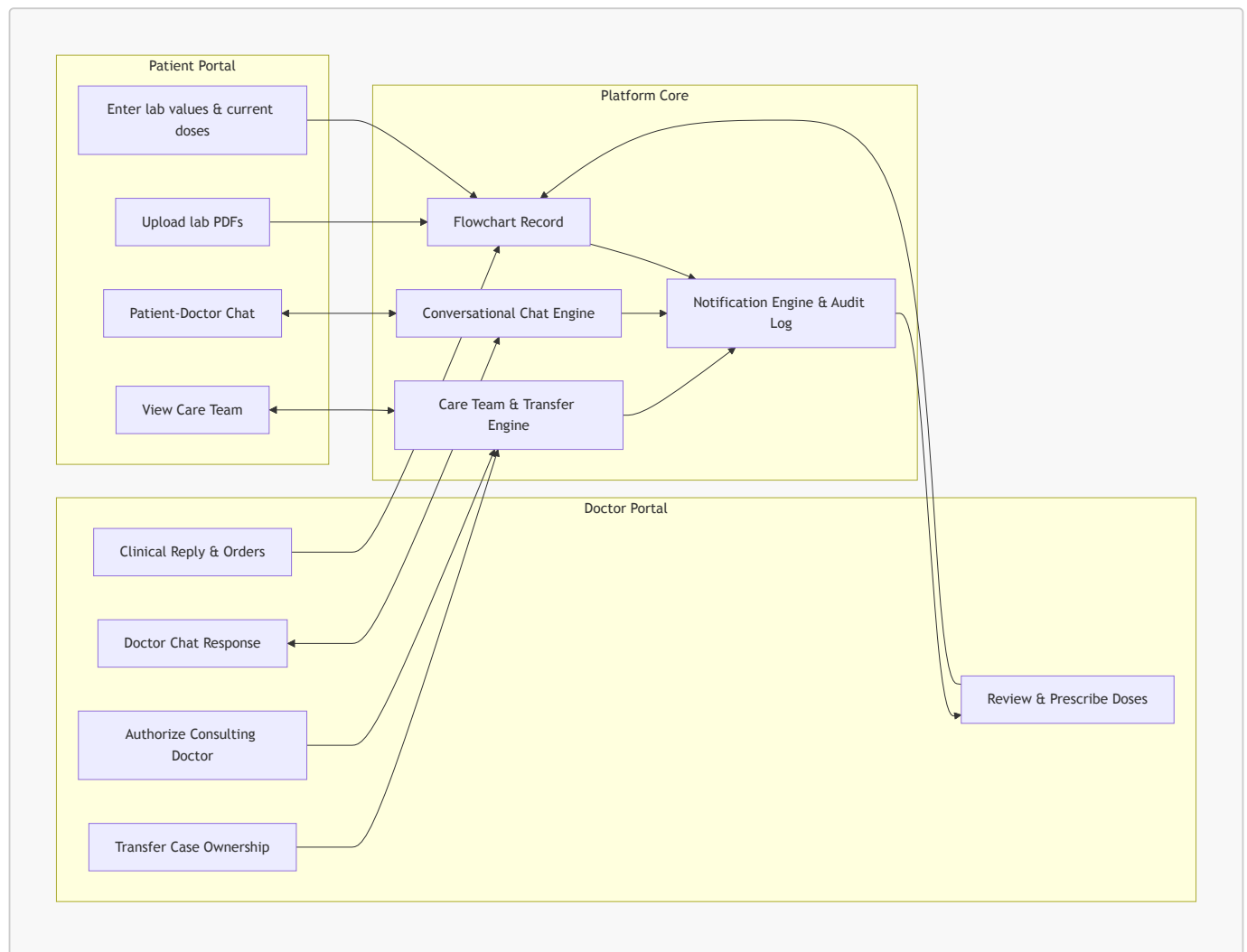
- Multiple dated rows track progress (e.g., 13/6/26 → 13/7/26).
- Lab columns cover hematology, liver function, renal function, electrolytes, and drug levels.
- Medication dose columns: **Neoral/Tac, Everolimus, Aza/MPA, Pred.**
- Drug level columns: **Tac level, EVO level** (Everolimus level).
- Dose adjustments are visually distinguished (e.g., red ink: 4/4 → 2/2).
- Patient header includes diagnosis, surgery date, histopathology, and anastomosis details.
- Footer instructs: "Enter further investigation reports in above table and send to drrajeshdey@gmail.com".

6. Proposed Solution (To-Be)

A **web/mobile application** featuring:

1. A **Patient portal** — digital flowchart, row entry, lab upload, conversational chat, care team view.

2. A **Doctor portal** — dashboard, chart review, dose entry, clinical reply, care team chat, authorization delegation, case transfer dialog.
3. A **Conversational Chat Engine** — 1:1 and team messaging between patient and authorized doctors, complete with voice notes, attachments, and automated system event posts.
4. A **Care Team & Transfer Authorization Engine** — role-based access control for primary, co-managing, consulting, and transferred doctors.
5. A **Notification layer** — alerts for lab submissions, doctor replies, chat messages, consult invitations, and case transfers.
6. An **Audit trail** — immutable history of labs, dose edits, chat transcripts, permission grants, and case handovers.



7. Functional Requirements

7.1 Patient profile & chart header

Each patient chart stores static header fields matching the paper form:

Field	Example from sample	Required
Patient name	Raghavendra S. Dyk	Yes
Age / sex	4.5 yrs / M	Yes
Hospital / Max ID	SHMS.750590	Yes

Field	Example from sample	Required
Date of operation	26/05/2026	Yes
Diagnosis	DCLD - ? AIH	Yes
Histopathology	Biliary Cirrhosis (PBC)	Yes
Type of biliary anastomosis	Free text	Yes
Patient photo	Passport-size image	Optional
Primary Doctor	Dr. Rajesh Dey	Yes
Authorized Care Team	Dr. Tejai B (Co-Managing), Dr. Barun (Consultant)	Yes
Primary reporting email	drrajeshdey@gmail.com	Yes

7.2 Follow-up row (investigation table)

Each follow-up is one **dated row** in the flowchart. All columns below are captured on every submission unless marked optional.

7.2.1 Complete field catalog

The following table is the **authoritative list** of flowchart fields. UI labels may use clinic-preferred abbreviations; the system stores values using the canonical field key.

#	UI label	Canonical key	Category	Type	Unit / format	Required
—	PP Date	pp_date	Meta	Date	DD/MM/YYYY	Yes
1	Hb	hb	Lab	Numeric	g/dL	No
2	TLC	tlc	Lab	Numeric	×10 ³ /μL	No
3	PLT	plt	Lab	Numeric	×10 ³ /μL	No
4	AFP	afp	Lab	Numeric	ng/mL	No
5	INR	inr	Lab	Numeric	ratio	No
6	PTT	ptt	Lab	Numeric	seconds	No
7	Bilirubin Total	bilirubin_total	Lab	Numeric	mg/dL	No
8	Bilirubin Direct	bilirubin_direct	Lab	Numeric	mg/dL	No
9	SGOT	sgot	Lab	Numeric	U/L	No
10	SGPT	sgpt	Lab	Numeric	U/L	No
11	ALK PHOS	alk_phos	Lab	Numeric	U/L	No
12	GGT	ggt	Lab	Numeric	U/L	No
13	Albumin	albumin	Lab	Numeric	g/dL	No

#	UI label	Canonical key	Category	Type	Unit / format	Required
14	Na/K	na_k	Lab	Text or split numeric	e.g. 136/4.2	No
15	Urea	urea	Lab	Numeric	mg/dL	No
16	Creatinine	creatinine	Lab	Numeric	mg/dL	No
17	HbA1c	hba1c	Lab	Numeric	%	No
18	Tac Level	tac_level	Drug level	Numeric	ng/mL (trough C0)	No
19	EVO level	evo_level	Drug level	Numeric	ng/mL	No
20	Neoral/Tac	neoral_tac	Medication dose	Dose notation	e.g. 4/4	No
21	Everolimus	everolimus	Medication dose	Dose notation	e.g. 1/0	No
22	Aza/MPA	aza_mpa	Medication dose	Dose notation	e.g. 1/1	No
23	Pred	pred	Medication dose	Dose notation	e.g. 5 or 5/0	No
24	Wt	wt	Clinical	Numeric	kg	No
25	Comments	comments	Notes	Text	Free text per row	No

Legacy / alias mapping (paper form and earlier spec versions):

Legacy label	Maps to
Platelet count	PLT (plt)
Bil Total	Bilirubin Total (bilirubin_total)
Alb	Albumin (albumin)
Creat	Creatinine (creatinine)
Tac/C0 level, Tec Level	Tac Level (tac_level)
Everolimus level	EVO level (evo_level)
Tac/Cyclo, Neoral, Tacrolimus/Cyclosporine	Neoral/Tac (neoral_tac)
Wys, Wysolone, Prednisolone	Pred (pred)

7.2.2 Lab columns

All lab fields are optional on each row (partial entry allowed). Numeric validation applies where applicable.

Column	Canonical key	Notes
Hb	hb	Hemoglobin

Column	Canonical key	Notes
TLC	<code>tlc</code>	Total leukocyte count
PLT	<code>plt</code>	Platelet count
AFP	<code>afp</code>	Alpha-fetoprotein
INR	<code>inr</code>	International normalized ratio
PTT	<code>ptt</code>	Partial thromboplastin time
Bilirubin Total	<code>bilirubin_total</code>	Total bilirubin
Bilirubin Direct	<code>bilirubin_direct</code>	Direct (conjugated) bilirubin
SGOT	<code>sgot</code>	AST
SGPT	<code>sgpt</code>	ALT
ALK PHOS	<code>alk_phos</code>	Alkaline phosphatase
GGT	<code>ggt</code>	Gamma-glutamyl transferase
Albumin	<code>albumin</code>	Serum albumin
Na/K	<code>na_k</code>	Sodium/potassium — single field or split <code>na + k</code>
Urea	<code>urea</code>	Blood urea
Creatinine	<code>creatinine</code>	Serum creatinine
HbA1c	<code>hba1c</code>	Glycated hemoglobin

7.2.3 Drug level columns

Column	Canonical key	Notes
Tac Level	<code>tac_level</code>	Tacrolimus trough (C0); supersedes legacy labels Tac/C0 level, Tec Level
EVO level	<code>evo_level</code>	Everolimus serum level; alias Everolimus level

7.2.4 Medication dose columns

Patient-entered on submit; doctor may override on review. Dose notation format to be confirmed with clinic (see §17 open questions).

Column	Canonical key	Notes
Neoral/Tac	<code>neoral_tac</code>	Combined tacrolimus/cyclosporine (Neoral) dose; legacy label Tac/Cyclo
Everolimus	<code>everolimus</code>	Everolimus dose (distinct from EVO level)
Aza/MPA	<code>aza_mpa</code>	Azathioprine or mycophenolate dose
Pred	<code>pred</code>	Prednisolone / Wysolone dose; legacy label Wys

7.2.5 Weight, comments, and attachments

Column	Canonical key	Notes
Wt	wt	Body weight in kg
Comments	comments	Free-text notes for the row (patient or doctor); visible in chart and exports
Attachments	attachments[]	Lab report PDFs/images (see §7.3 P-03); not a flowchart column but linked to the row

Business rules:

- On **patient submission**, the patient fills lab values, drug levels, **current medication doses**, weight, and optional **Comments**.
- On **doctor review**, medication dose columns may appear **blank** until the doctor enters **prescribed doses** for the next interval; doctor may add or amend **Comments**.
- Doctor-entered doses must be visually distinct from patient-reported doses (replacing "red ink").
- Support **dose change history** per medication column (old value → new value, timestamp, prescribing doctor).
- **Comments** are append-only in the audit log when edited after review (original text retained).

7.3 Patient portal features

ID	Requirement	Priority
P-01	View full digital flowchart (read-only history + editable new row)	Must
P-02	Add a new follow-up row supporting all lab, drug level, and medication dose fields	Must
P-03	Upload lab report files (PDF/image) linked to a follow-up date	Must
P-04	Submit follow-up for doctor review	Must
P-05	View doctor's latest response (prescribed doses, additional tests, notes)	Must
P-06	Receive notifications when doctor responds or messages in chat	Must
P-07	View dose change highlights on the chart	Must
P-08	Conversational Chat: Send text, voice notes, and image attachments to care team	Must
P-09	Care Team View: View current primary doctor and authorized co-managing/consulting doctors	Must

7.4 Doctor portal features

ID	Requirement	Priority
D-01	Dashboard of patients with pending submissions and unread chat messages	Must

ID	Requirement	Priority
D-02	Open patient chart with tabular history and trend view	Must
D-03	Review patient-entered labs, drug levels, and current doses	Must
D-04	Enter prescribed doses in medication columns for the reviewed visit	Must
D-05	Send structured response to patient	Must
D-06	Free-text clinical reply (additional medications, lifestyle advice, etc.)	Must
D-07	Request additional investigations (structured checklist + notes)	Must
D-08	View uploaded lab reports inline or downloadable	Must
D-09	See prior dose changes and who made them	Must
D-10	Conversational Chat: Reply to patient chat queries, send voice notes, attach files	Must
D-11	Care Team Delegation: Authorize secondary doctors with specific permission levels	Must
D-12	Case Transfer: Initiate and execute patient case transfer to another primary doctor	Must

7.5 Notifications & delivery

Channel	Use case	Priority
In-app	Submission received, doctor responded, new chat message, case transfer notification	Must
Email	Official record delivery, consult invitation, case handoff summaries	Must
SMS	Milestone reminders, urgent chat alerts, OTP	Must
WhatsApp	Milestone reminders, doctor responses, chat message alerts	Must
Push (mobile)	Real-time chat messages, time-sensitive doctor replies	Must

7.6 Milestone reminders (SMS & WhatsApp)

(Retained as specified in Section 7.6).

7.7 Reporting & export

ID	Requirement	Priority
R-01	Export chart as PDF matching paper layout	Should
R-02	Print-friendly flowchart view	Should
R-03	Export audit log of dose changes, chat transcripts, and case transfer history	Could

7.8 Conversational Chat (Doctor–Patient & Care Team)

The platform provides a secure, HIPAA/PHI-compliant messaging hub replacing informal WhatsApp/email communication.

7.8.1 Chat Scope & Features

ID	Requirement	Priority
C-01	Patient-Care Team Thread: Each patient chart has an active main chat thread connecting the patient/caregiver with all authorized doctors on their care team.	Must
C-02	Multi-Format Messages: Support text messages, image attachments (photos of symptoms/incision), document attachments (PDFs), and recorded voice notes .	Must
C-03	Automated Clinical System Messages: System posts automated event cards into the chat thread when key actions occur (e.g. <i>"Patient submitted labs for 23/08/2026"</i> , <i>"Dr. Dey updated Pred dose to 2.5mg"</i> , <i>"Case transferred to Dr. Tejai B"</i>).	Must
C-04	Contextual Item Linking: Users can quote or link a specific message to a follow-up row ID, lab value, or dose change for clinical context.	Must
C-05	Read Receipts & Status Indicators: Show message status (Sent, Delivered, Read by Doctor / Read by Patient) with timestamps.	Must
C-06	Non-Emergency Disclaimer Banner: Permanent banner at top of chat: <i>"Chat is for non-urgent follow-up queries only. In case of medical emergency, contact ER immediately."</i>	Must
C-07	Urgency Triage Flag: Patient can mark a message as "Routine Query" or "Symptom Concern"; symptom concerns highlight in doctor dashboard.	Should
C-08	Doctor-to-Doctor Internal Notes / Consult Thread: Secondary thread on the same patient chart visible ONLY to authorized doctors (hidden from patient) for inter-specialist case discussions.	Should
C-09	Search & Filter: Search chat history by key terms or filter by media/attachments.	Could
C-10	Immutable Audit Log: Chat transcripts cannot be edited or deleted by users; retained as part of permanent electronic health record.	Must

7.9 Multi-Doctor Participation, Authorization & Case Transfer

Post-operative care frequently requires collaboration between multiple doctors or transferring patient ownership due to shifts, sabbaticals, or specialty handovers.

7.9.1 Doctor Roles & Access Hierarchy

Role	Scope of Access & Permissions
Primary Doctor	Full control: view chart, prescribe/adjust doses, order labs, reply & chat with patient, invite/authorize secondary doctors, modify access levels, initiate case transfer.
Co-Managing Doctor	Full clinical care access: view chart, prescribe/adjust doses, order labs, reply & chat with patient. <i>Cannot authorize new doctors or transfer case ownership.</i>

Role	Scope of Access & Permissions
Consulting Doctor (View & Note)	Specialist access: view chart, read patient chat, post clinical advice in inter-doctor consult thread. <i>Cannot adjust doses or prescribe directly unless authorized.</i>
Transferred (Former) Primary Doctor	Historical read-only access: view chart entries and chat logs recorded <i>up to the timestamp of transfer</i> . No access to post-transfer entries unless re-invited.

7.9.2 Authorization & Delegation Requirements

ID	Requirement	Priority
T-01	Grant Doctor Access: Primary Doctor (or Admin) can invite another registered doctor to join the patient's care team by email or doctor ID.	Must
T-02	Access Level Specification: Primary Doctor selects access role (Co-Managing vs Consulting View-Only) when adding a doctor.	Must
T-03	Time-Bound Access: Primary Doctor can optionally set an expiration date for consulting access (e.g., 14-day consult).	Should
T-04	Revoke / Modify Access: Primary Doctor can revoke access or adjust permissions of any secondary doctor at any time.	Must
T-05	Patient Care Team Visibility: Patient portal displays active care team members with their names, photos, and clinical roles.	Must
T-06	Patient Notification: Patient is notified via app/SMS/WhatsApp whenever a new doctor is authorized or added to their care team.	Must

7.9.3 Patient Case Transfer Requirements

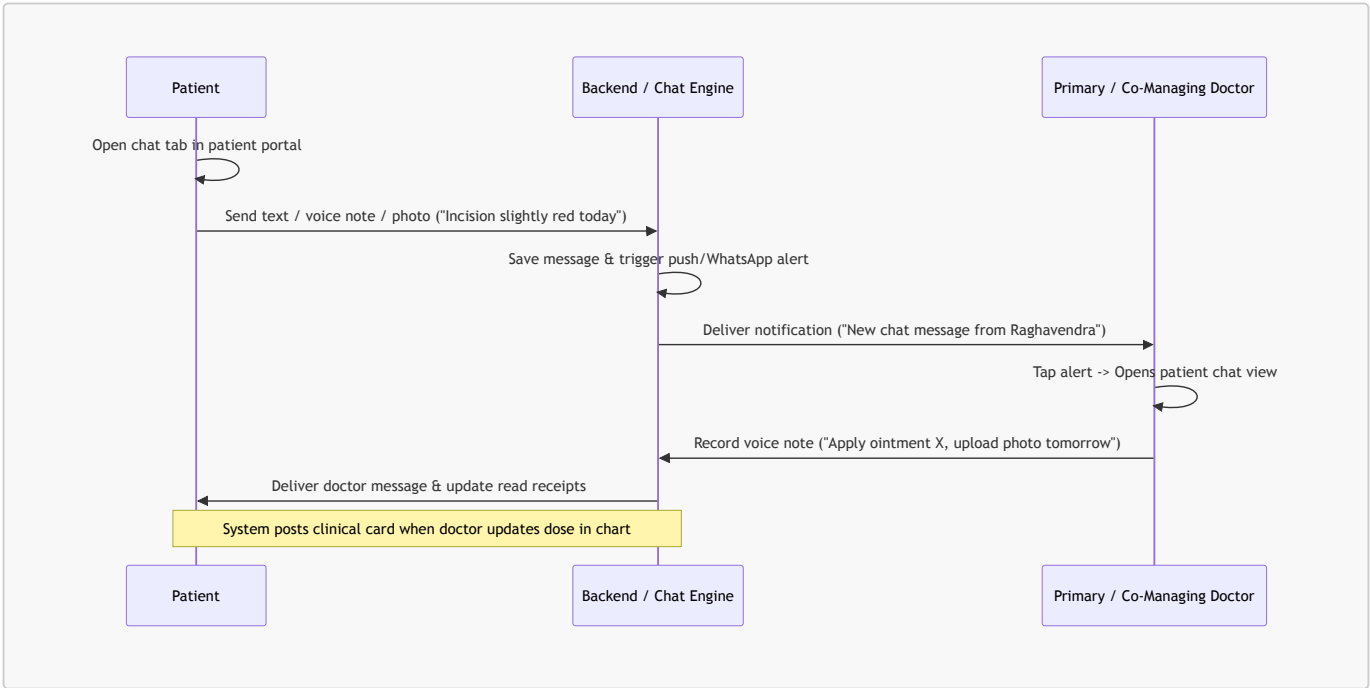
ID	Requirement	Priority
T-07	Initiate Transfer: Primary Doctor can select a target doctor and initiate a primary ownership transfer request.	Must
T-08	Handoff Summary: Transfer dialog requires a structured handoff note (reason for transfer, current clinical status, key warnings).	Must
T-09	Transfer Acceptance Workflow: Target doctor receives handoff notification and must accept (or decline with reason) to finalize transfer.	Must
T-10	Admin Override Transfer: Clinic Admin can execute an immediate transfer without target acceptance (e.g. emergency doctor unavailability).	Must
T-11	Ownership Transition: Upon transfer completion, target doctor becomes Primary Doctor ; former doctor transitions to Transferred (Read-Only) or Co-Managing based on transfer settings.	Must
T-12	Patient & Team Notification: System broadcasts automated notifications to patient, incoming primary doctor, outgoing primary doctor, and care team.	Must
T-13	System Chat Post: Automated system message posted into patient chat documenting transfer: <i>"Primary care transferred from Dr. Rajesh Dey to Dr. Tejai B on [date]"</i> .	Must

ID	Requirement	Priority
T-14	Transfer Audit Trail: Log full history of case transfers (from_doctor, to_doctor, initiated_by, handoff_note, accepted_at) in immutable audit log.	Must

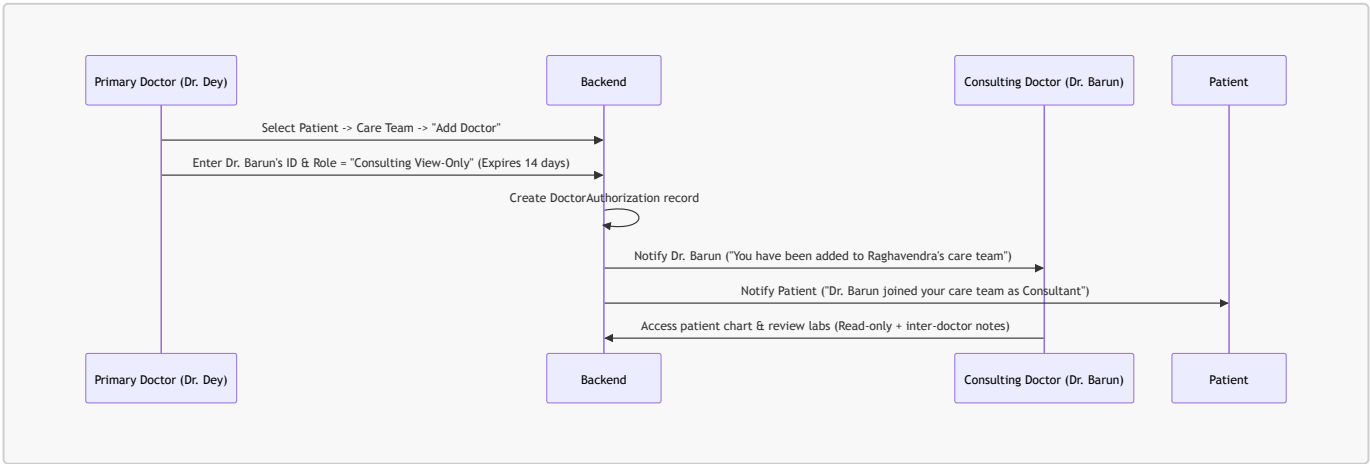
8. User Flows

(Flows 8.1 to 8.4 retained from previous spec).

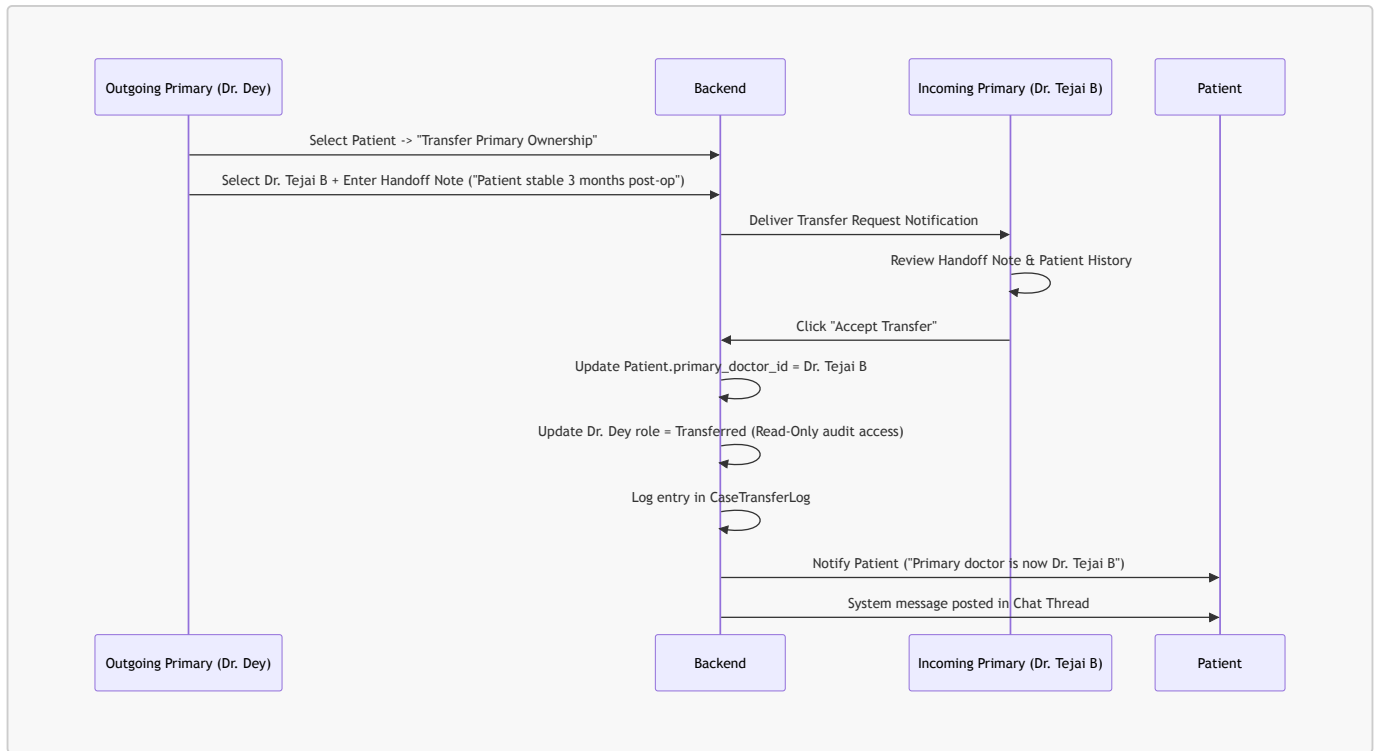
8.5 Conversational Chat Flow



8.6 Multi-Doctor Authorization Flow



8.7 Patient Case Transfer Flow



9. Data Model (Conceptual)

Patient

- id, name, age, sex, max_id, photo_url
- date_of_operation, diagnosis, histopathology, anastomosis_type
- primary_doctor_id (FK to User where role=doctor)
- assigned_doctors[] (computed list of active authorized doctors)
- contact_email, phone_number, whatsapp_number
- reminder_preferences { sms_enabled, whatsapp_enabled, timezone }

DoctorAuthorization

- id, patient_id, primary_doctor_id
- authorized_doctor_id (FK to User)
- access_level [co_managing | consult_view]
- granted_by (user_id), granted_at
- expires_at (optional)
- status [active | revoked | expired]
- revoked_at, revoked_by

CaseTransferLog

- id, patient_id
- previous_primary_doctor_id
- new_primary_doctor_id
- initiated_by (user_id)
- handoff_note (text)
- status [pending | accepted | declined | admin_forced]
- initiated_at, responded_at
- decline_reason (optional)

ChatThread

- id, patient_id
- thread_type [patient_care_team | doctor_internal_consult]
- title

```
└─ created_at, updated_at
└─ last_message_at
```

ChatMessage

```
└─ id, thread_id, sender_id (user_id)
└─ sender_role [patient | caregiver | primary_doctor | co_managing_doctor |
consulting_doctor | system]
└─ message_type [text | voice_note | attachment | clinical_event_card]
└─ content (text body or transcribed voice note)
└─ voice_note_url, voice_duration_seconds
└─ linked_follow_up_row_id (optional FK)
└─ linked_dose_change_id (optional FK)
└─ urgency_flag [routine | symptom_concern]
└─ read_by_user_ids[]
└─ sent_at
```

ChatAttachment

```
└─ id, message_id
└─ file_url, file_type [image | pdf | audio]
└─ file_name, file_size_bytes
└─ uploaded_at
```

FollowUpRow

```
└─ id, patient_id, pp_date, status [draft | pending | reviewed]
└─ lab_values { hb, tlc, plt, afp, inr, ptt, bilirubin_total, bilirubin_direct,
sgot, sgpt, alk_phos, ggt, albumin, na_k, urea, creatinine, hba1c }
└─ drug_levels { tac_level, evo_level }
└─ patient_reported_doses { neoral_tac, everolimus, aza_mpa, pred }
└─ doctor_prescribed_doses { neoral_tac, everolimus, aza_mpa, pred }
└─ wt, comments, attachments[]
└─ submitted_at, reviewed_at
└─ reviewed_by (doctor_id)
```

DoseChange

```
└─ follow_up_row_id, field_name, old_value, new_value
└─ changed_by (doctor_id), changed_at, reason
```

DoctorResponse

```
└─ follow_up_row_id, doctor_id, additional_tests[], additional_medications,
clinical_notes, sent_at
```

User

```
└─ id, role [patient | caregiver | doctor | admin]
└─ display_name, email, phone, medical_license_number (for doctors),
profile_photo_url
└─ status [active | disabled]
```

10. UI/UX Requirements

10.1 Chat Interface Design

- **Patient view:** Clean 1:1 style messaging UI showing messages from all care team members with clear doctor name & avatar labels. Dedicated voice record button and file upload clip.

- **Doctor view:** Integrated chat sidebar or tab next to patient chart. Includes filter for "Internal Doctor Notes" vs "Patient Chat". System event cards rendered in distinct neutral style.

10.2 Care Team & Authorization UI

- **Care Team Panel:** Modal/tab on patient chart displaying Primary Doctor and authorized co-managing/consulting doctors.
- **Add Doctor Dialog:** Doctor lookup search, role radio buttons (Co-Managing vs Consulting View-Only), optional expiration picker.

10.3 Case Transfer Dialog

- **Transfer Form:** Target doctor selector, handoff note text box, checkbox to keep outgoing doctor as co-managing or remove access.
- **Transfer Banner:** Incoming doctor sees prominent acceptance banner on dashboard: "Pending Case Transfer: Raghavendra S. Dyk from Dr. Rajesh Dey. [Review Handoff & Accept]".

11. Non-Functional Requirements

(Retained from previous spec, with addition of HIPAA/PHI compliant encrypted chat message storage and web socket real-time messaging latency < 1s).

12. Authentication & Authorization

12.10 Granular Authorization Matrix

User Role	View Chart History	Enter Patient Labs/Doses	Prescribe/Adjust Doses	Send Patient Chat	Post Doctor Internal Notes	Authorize Secondary Doctor	Transfer Case Ownership
Patient / Caregiver	Own chart only	Yes (New row)	No	Yes	No	No	No
Primary Doctor	Full	Yes	Yes	Yes	Yes	Yes	Yes
Co-Managing Doctor	Full	Yes	Yes	Yes	Yes	No	No
Consulting Doctor	Full	No	No	Read-only	Yes	No	No
Transferred Doctor	Historical only	No	No	No	Read-only	No	No
Admin	Full (Audit)	No	No	No	No	Yes	Yes (Force)

13–14. Security, Compliance & Scope

- Phase 1 (MVP) includes: 1:1 Patient-Care Team Chat, Primary Doctor Authorization Grants, Primary Case Transfer, and Tabular Flowchart.
- Phase 2 includes: Voice Notes, Doctor-to-Doctor Internal Consult Thread, and Automated Audio Transcription.

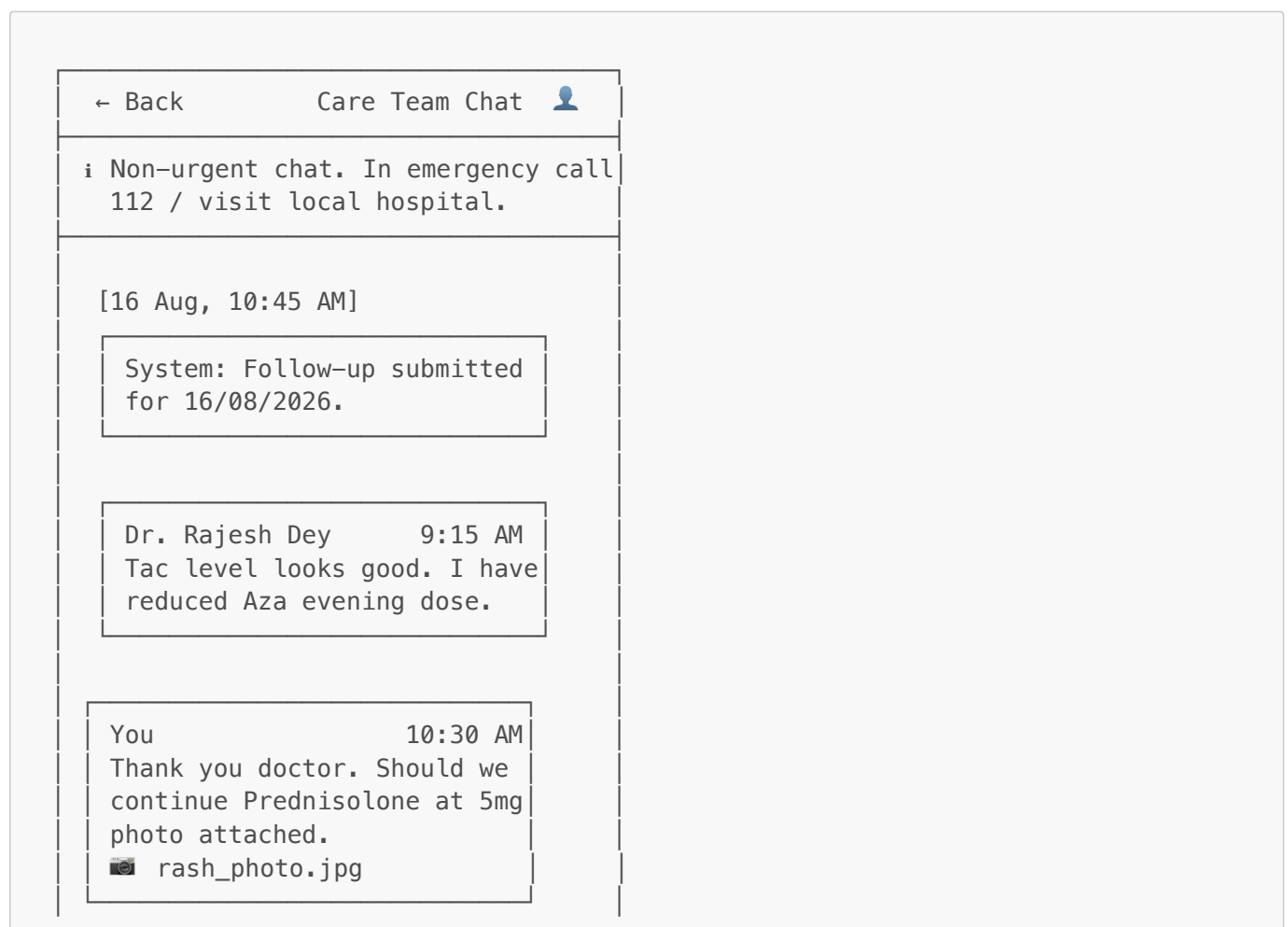
15. Sample Screen Inventory

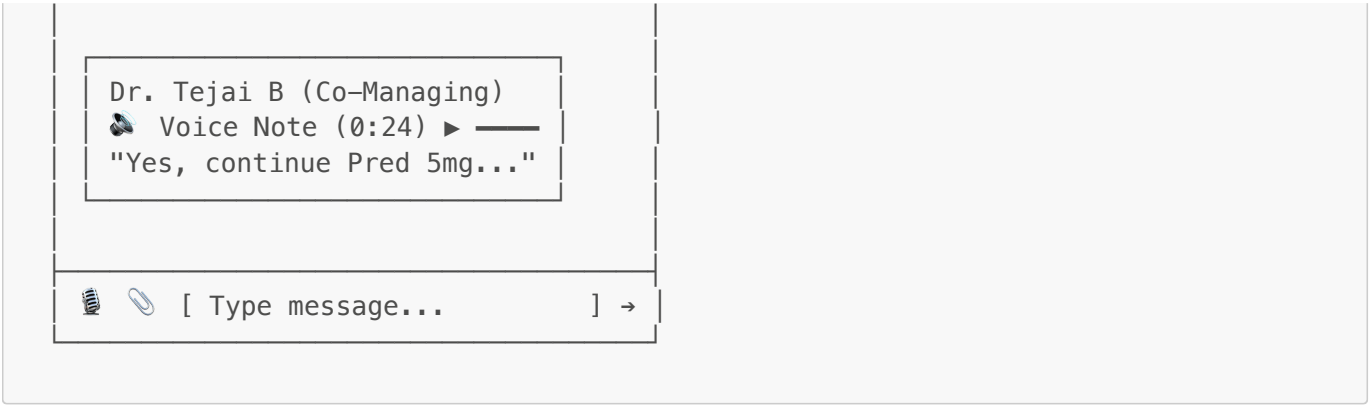
Screen	User	Purpose
My Chat	Patient	Message care team, record voice notes, send photos
Patient Care Team	Patient	View primary doctor and authorized consulting doctors
Doctor Chat Hub	Doctor	View patient chat, send voice notes, view system event posts
Internal Consult Thread	Doctor	Private doctor-to-doctor discussion for complex cases
Care Team Management	Doctor, Admin	Invite secondary doctors, assign access roles, revoke access
Case Transfer Modal	Doctor, Admin	Select incoming doctor, assign access roles, execute transfer
Transfer Review Banner	Doctor	Incoming doctor accepts/declines pending case transfer

16. Mock Screens

16.16 Patient — Conversational Chat

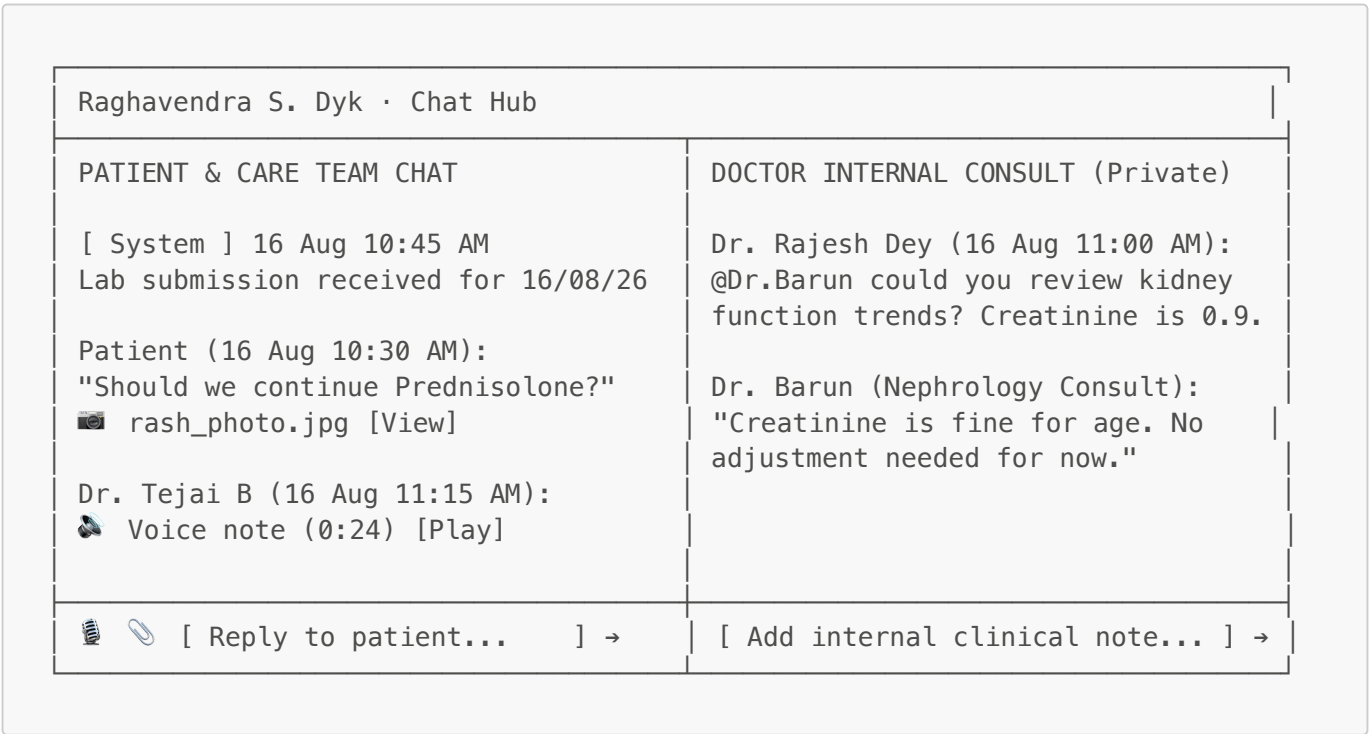
Route: </patient/chat>





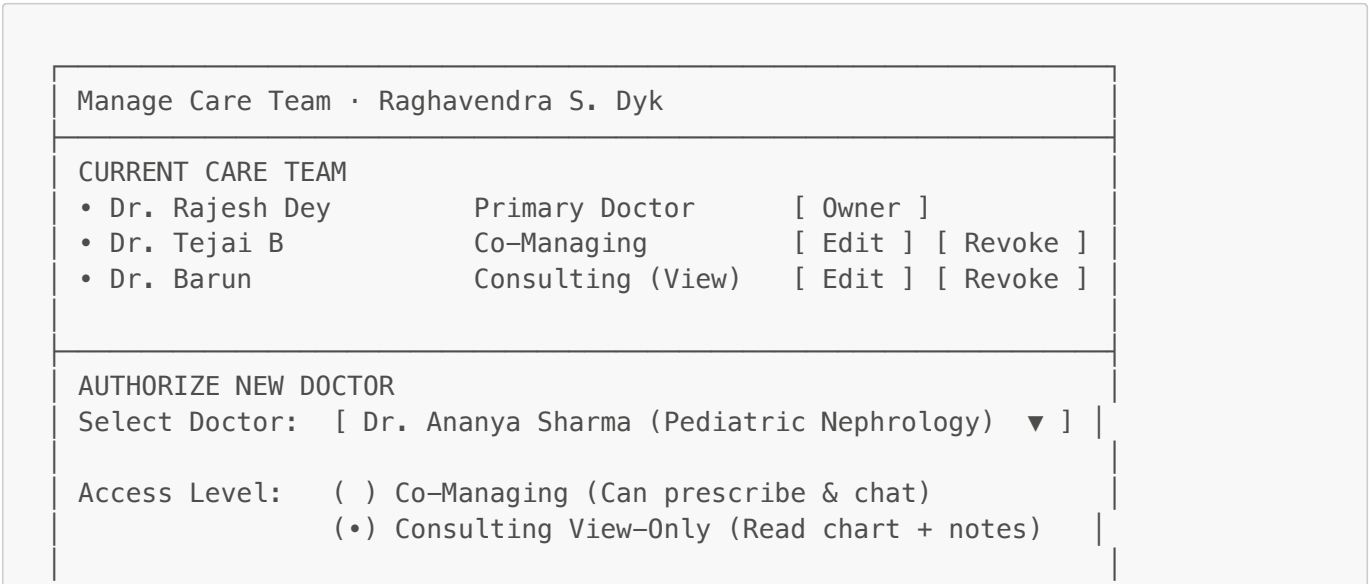
16.17 Doctor — Integrated Chat & Consult View

Route: /doctor/patients/:id/chat



16.18 Doctor — Care Team Management & Authorization Modal

Route: /doctor/patients/:id/care-team



Access Duration: [14 Days (Temporary Consult) ▼]

Grant Authorization

16.19 Doctor — Patient Case Transfer Modal

Route: `/doctor/patients/:id/transfer`

Transfer Primary Ownership · Raghavendra S. Dyk

Current Primary: Dr. Rajesh Dey

Select New Primary Doctor:

[Dr. Tejai B (Transplant Surgery) ▼]

Handoff Summary & Clinical Status (Required):

Patient is 3 months post liver transplant. Tac level stable at 8.1. Aza dose reduced. Transferring primary ownership as Dr. Tejai B takes over post-op clinic rotations.

Outgoing Doctor Access:

[X] Retain Dr. Rajesh Dey as Co-Managing Doctor

[] Transition Dr. Rajesh Dey to Historical Read-Only

Initiate Primary Ownership Transfer

17. Open Questions for Stakeholders

1. **Dose notation:** What exactly does 4/4 mean for Neoral/Tac — AM/PM, brand split, or mg split? Document canonical format.
2. **Column set:** Is the column list fixed for all patients, or configurable by diagnosis/protocol?
3. **Immunosuppression regimen:** Which patients use Everolimus dose + EVO level vs Neoral/Tac + Tac level only? Are both ever required on the same row?
4. **Who can submit:** Patient directly, caregiver only, or clinic staff on behalf of patient?
5. **WhatsApp templates:** Which pre-approved WhatsApp message templates are required by the clinic's API provider?
6. **Legal/regulatory:** Is this clinic-operated software or a regulated medical device in your jurisdiction?
7. **Emergency handling:** Should the app include escalation if critical lab values are entered?
8. **Multi-language:** Hindi/regional language support needed?
9. **SMS provider:** Which SMS gateway will be used for OTP delivery in production (MSG91, Twilio, etc.)?
10. **Doctor onboarding:** Should doctors self-register via Google/X/Facebook, or admin-provision only?

11. **WhatsApp provider:** Which WhatsApp Business API partner will be used (Meta Cloud API, Gupshup, etc.)?
 12. **Default protocol:** What are the standard post-op checkpoint days and follow-up interval for this clinic (e.g., 7/14/30/90 days)?
 13. **Overdue escalation:** Should the assigned doctor be notified automatically when a patient misses a milestone?
 14. **Voice Note Duration Limit:** Should voice recording be capped at 60 or 120 seconds to manage bandwidth and storage?
 15. **Transfer Acceptance Policy:** Should primary case transfer take effect immediately upon initiation or require explicit click-to-accept by incoming doctor? (Recommendation: Require acceptance with admin override).
 16. **Emergency Escalation:** Should high urgency chat flags send direct SMS/push alerts to the doctor on duty?
-

18. Acceptance Criteria (MVP)

The MVP is accepted when:

1. A patient can complete and submit a follow-up row that includes all fields defined in §7.2.1 (Hb through Comments).
2. A doctor can open the submission, enter prescribed doses, and send a reply with additional test requests.
3. The patient receives the doctor's response without using manual email for that transaction.
4. Dose changes are visibly distinguished and stored in an audit log.
5. The chart can be exported/printed in a layout recognizable to users of the paper form.
6. Only authorized users can access a given patient's PHI.
7. Users can sign in via **Google OAuth**, **X OAuth**, **Facebook Login**, or **SMS OTP** and reach the correct portal for their role.
8. Failed logins, OTP attempts, and sign-outs are recorded in the authentication audit log.
9. Milestones are auto-generated from the post-op protocol and rolling follow-up interval.
10. Patients receive automated **SMS and/or WhatsApp** reminders before and on milestone due dates, per their preferences.
11. Submitting a follow-up cancels pending reminders for that milestone; delivery status is logged for each reminder sent.
12. Patient and authorized doctors can exchange real-time text, voice notes, and image attachments in the patient chat thread.
13. System automatically posts clinical event cards in the chat whenever lab rows are submitted or dose changes are made.
14. Primary doctor can grant co-managing or consulting access to secondary doctors, and patient portal updates the care team display accordingly.
15. Primary doctor can initiate a case transfer with a mandatory handoff note, and upon acceptance, primary ownership is updated with complete audit log entries.
16. All flowchart entries, dose adjustments, notifications, and milestone reminders operate as specified in Sections 7–16.